Kamil Belán

Curriculum vitae

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kamilbelan

Education

2026–present **Ph.D.**, *Faculty of Mathematics and Physics, Charles University*, Prague, Computational Mathematics

2024–2026 **M.Sc.**, *Charles University*, Prague, Mathematical and Computational Modelling in Physics

2021–2024 **B.Sc.**, *Charles University*, Prague, Physics

Research interests

Numerical methods for partial differential equations of fluid dynamics; past work includes

- stratified compressible atmospheric flow
- Lagrangian (particle) methods, in particular Smoothed Particle Hydrodynamics (SPH)
- well-balanced numerical schemes

Current focus: high-order methods, in particular

- discontinuous Galerkin method (DGM)
- *hp*-adaptive discontinuous Galerkin method
- space-time discontinuous Galerkin method (STDGM)

Theses

2026 **Numerical methods for the simulation of air flow past a mountain**, *master's thesis*, supervisor Assoc. Prof. Michal Pavelka

The objective of the thesis is to further develop and verify the SPH method for compressible atmospheric flow over orography. The work relies on a reformulation of the problem in the potential temperature variable, robust terrain-following orography description and improved treatment of inflow and outflow boundary conditions. The numerical instability of the hydrostatic balance is addressed by the development of novel well-balanced schemes and adaptive techniques.

Codebase (both theses) `https://github.com/kamilbelan/sph-mountain-waves`

2024 **Simulating flows past a mountain range using smoothed particle hydrodynamics**, *bachelor's thesis*, supervisor Assoc. Prof. Michal Pavelka

The main objective of the thesis is to use the Lagrangian numerical method SPH to simulate compressible atmospheric flow over orography, a problem typically solved by mesh-based methods (FVM, FDM). It was shown by numerical evidence that the simulation of hydrostatic balance of a stratified atmosphere by the standard SPH method is so unstable that SPH in its standard formulation is completely unable to capture phenomena such as internal gravity waves.

Technical skills

Programming languages: Julia, Python, Mathematica, MATLAB

Tools: LaTeX, Git, SSH, bash, Slurm, Docker

Languages

Czech: native

English: **C1** – Cambridge certificate

German: ~**B1** – courses at Charles University